

AI-Powered Recommendation Engine for Indian Standards (IS) in Procurement
Implementation Plan

1. Problem Framing & Scope

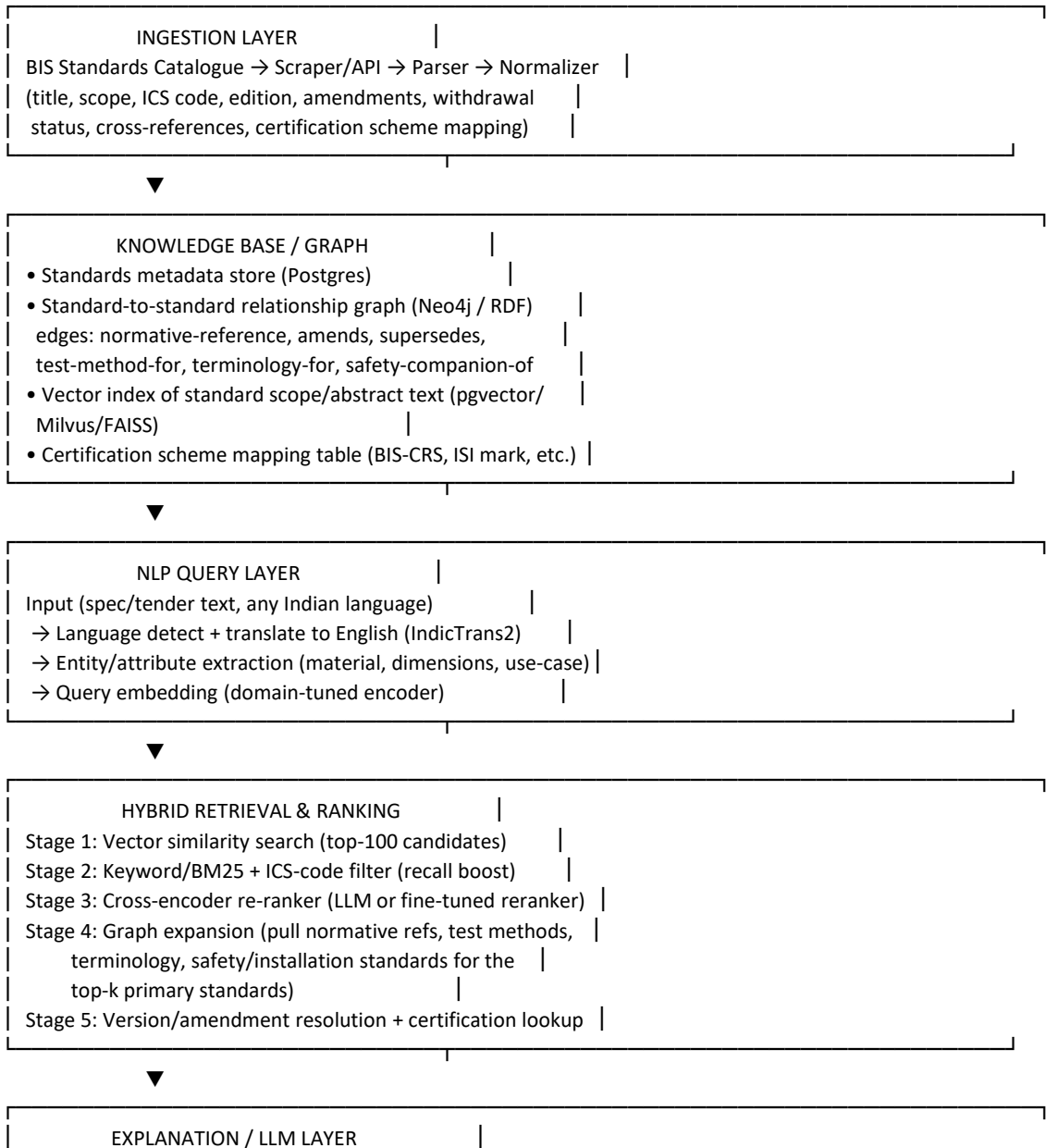
Core task: Given a free-text product description / technical specification / tender document (possibly in a regional language), return a ranked list of applicable Indian Standards, plus allied/normative/test-method/safety/installation standards, current version status, and applicable certification requirements (BIS, CRS, Hallmarking, etc.).

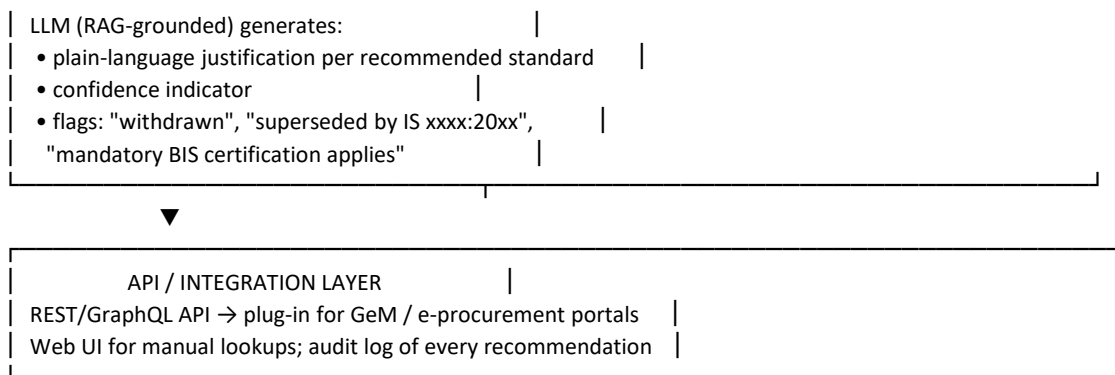
This is fundamentally a semantic retrieval + knowledge-graph reasoning problem, not a classification problem — the standards corpus is large, overlapping, and constantly revised, so the system must retrieve, rank, and *explain* rather than just classify into fixed buckets.

Two engines are needed working together:

- 1. Retrieval Engine — finds candidate IS standards semantically relevant to the input.
- 2. Standards Graph Engine — expands the top candidates into allied/normative/referenced standards and attaches certification & version metadata.

2. High-Level Architecture





3. Data Foundation (the hardest and most critical part)

Data need	Source strategy
Master list of IS standards, titles, scope, ICS classification	BIS Standards Catalogue / BIS e-Standards portal (licensed bulk data or structured scraping with permission)
Full text of standards (scope, normative references section)	BIS document purchase/API where available; at minimum scope + reference list is essential
Amendment & revision history, withdrawal status	BIS "Status" pages / periodic BIS Gazette notifications
Cross-reference graph (which standards cite which)	Parsed from "Normative References" section of each standard
Certification scheme mapping (mandatory BIS certification list, CRS list, Hallmarking)	BIS mandatory certification list (QCO — Quality Control Orders), Ministry notifications
Product/tender description examples for training & eval	Historical GeM tender specs, CPWD/PWD schedules, past procurement documents (anonymized)

Key risk: BIS does not provide full standard text openly (it's a paid catalogue). Plan must budget for either a licensing arrangement with BIS or scoping the MVP to metadata (title/scope/ICS/cross-refs) rather than full text — this materially changes semantic quality and must be resolved in Phase 0.

4. Phased Delivery Plan

Phase 0 — Discovery & Data Access (3–4 weeks)

- Secure data-sharing agreement with BIS (or use whatever public catalogue fields are legally scrapable).
- Define standard schema: {IS_number, title, scope_text, ICS_code, edition_year, status, amendments[], normative_refs[], certification_scheme}.
- Collect a labeled evaluation set: ~500–1000 real tender/product descriptions manually mapped to correct IS numbers by domain experts (procurement officers + BIS technical committee input). This gold set drives every later quality decision.

Phase 1 — Knowledge Base Construction (4–6 weeks)

- Build ETL pipeline to ingest and normalize standards metadata into Postgres.
- Parse normative-reference sections to build the graph (Neo4j, or Postgres adjacency table if graph queries stay shallow).
- Build certification-mapping table from QCO notifications.
- Stand up a version-resolution service (given any IS number/edition, return latest valid edition + amendment chain).

Phase 2 — Retrieval MVP (4–5 weeks)

- Generate embeddings for standard titles+scope using a domain-adapted sentence embedding model (start with a strong general multilingual embedding model, e.g. an open multilingual E5/BGE-class model; fine-tune later once labeled data exists).
- Implement hybrid retrieval: vector search + BM25/ICS-code filter, fused via reciprocal rank fusion.

- Build a simple query API and internal test UI.
- Evaluate against the gold set using Recall@5/Recall@10 and MRR before moving on — do not proceed to re-ranking until baseline recall is acceptable (target Recall@10 $\geq$ 0.85 as an internal gate).

Phase 3 — Re-ranking, Graph Expansion & Explanation (4–6 weeks)

- Add cross-encoder or LLM-based re-ranker on top-100 candidates for precision.
- Implement graph-expansion logic: for each top-k primary standard, pull linked normative references, test-method standards, terminology standards, safety/installation standards, tagged by relationship type.
- Add LLM explanation layer (RAG over the retrieved scope text) to generate justifications and flag version/certification issues — constrain the LLM strictly to retrieved context to avoid hallucinated standard numbers (critical: fabricated IS numbers are a real risk and must be actively guarded against, e.g. by validating every cited IS number against the knowledge base before returning it).

Phase 4 — Multilingual & Tender-Document Input (3–4 weeks)

- Add language detection + translation (IndicTrans2 or equivalent) for regional-language queries.
- Add tender-document ingestion: PDF/DOCX parsing, section segmentation (extract only the technical-specification portion of a long tender document rather than feeding the whole file).

Phase 5 — Integration & Portal Plug-in (4–6 weeks)

- Build REST/GraphQL API with authentication, rate limiting, and audit logging (every recommendation + the official's final choice should be logged for continuous evaluation).
- Build a lightweight widget/plugin for e-procurement portals (e.g. GeM-style spec-builder integration) that suggests standards inline as the official types.
- Build a standalone web UI for ad hoc lookups.

Phase 6 — Pilot, Feedback Loop & Hardening (ongoing, 6+ weeks)

- Pilot with a small set of departments/PSEs on real tenders.
- Capture accept/reject/edit feedback on every recommendation → feed into re-ranker fine-tuning and gold-set expansion.
- Set up a scheduled job to detect newly withdrawn/superseded/amended standards and auto-flag any tenders currently referencing them.
- Security review, load testing, and accessibility review before wider rollout.

5. Core Technical Choices

Component	Recommendation	Why
Embedding model	Multilingual, domain-fine-tunable (e.g. E5/BGE-class open model)	Needed for semantic (not keyword) matching and multilingual input
Vector store	pgvector (if staying in Postgres) or Milvus for scale	Keeps ops simple at MVP scale; Milvus if corpus/query volume grows large
Graph store	Neo4j (or Postgres adjacency tables for MVP)	Standard-to-standard relationships are naturally graph-shaped
Re-ranker	Cross-encoder fine-tuned on gold set, or LLM-as-reranker	Precision boost after cheap recall-oriented retrieval
Generation/explanation	LLM via API, strictly RAG-grounded with post-hoc validation of cited IS numbers	Prevents hallucinated standard citations — non-negotiable for a compliance-facing tool
Translation	IndicTrans2 or equivalent open Indic MT model	Multilingual query support
Document parsing	Layout-aware PDF/DOCX parser (e.g. unstructured.io-style pipeline)	Tender documents are long and mixed-format

6. Evaluation Strategy

- Retrieval quality: Recall@5, Recall@10, MRR against the expert-labeled gold set — track separately for "primary standard identified" vs. "correct allied standards also surfaced."
- Version correctness: % of recommendations pointing to the current valid edition (not withdrawn/superseded) — should be near 100%, this is deterministic data-quality, not ML quality.
- Hallucination rate: % of cited IS numbers that fail validation against the knowledge base — target 0%, monitored continuously in production, not just at launch.
- Human-in-the-loop precision: % of top-3 recommendations accepted/used by procurement

officials during pilot, tracked per department.

- Latency: end-to-end response time target (e.g. <3–5s for interactive use in a spec-builder widget).

7. Key Risks & Mitigations

Risk	Mitigation
BIS full-text not freely available	Scope MVP to title/scope/ICS/cross-ref metadata; pursue formal data agreement with BIS in parallel
LLM hallucinates non-existent or wrong IS numbers	Hard validation layer: every IS number in the output must exist in the knowledge base and match the resolved current edition before being shown
Standards get revised/withdrawn after ingestion	Scheduled re-sync job against BIS status pages; version-resolution service is source of truth, not the vector index
Cold-start: little labeled query→standard data	Start with expert-curated gold set of a few hundred examples; expand via pilot feedback loop rather than trying to pre-label everything
Overlapping standard scopes cause ambiguous top-1 answers	Always return ranked top-k with confidence, not a single forced answer; let the human decide
Regional-language input mistranslation	Show both the translated query and original alongside results so officials can verify intent

8. Suggested Team & Timeline

- Team: 1 PM, 2 backend/data engineers, 1–2 ML/NLP engineers, 1 knowledge-graph/data engineer, 1 frontend engineer, 1 domain SME (ex-BIS/procurement background) part-time throughout, 1 QA.
- Total timeline to pilot-ready MVP: ~7–9 months across Phases 0–5, assuming data access is resolved early; Phase 6 (feedback hardening) continues indefinitely as a product operations function.

9. MVP Cut (if timeline is tight, e.g. hackathon-scale)

For a demo-scale MVP (weeks, not months):

1. Scrape/compile a few hundred IS standards' titles, scope, and ICS codes covering 2–3 product categories (e.g. cement, steel bars, electrical cables).
2. Build vector + keyword hybrid search over that subset.
3. Manually build a small cross-reference graph for those categories only.
4. Add an LLM layer to explain matches, with strict validation that any cited IS number exists in the local dataset.
5. Simple web UI: paste a spec → get ranked standards + allied standards + a certification flag if applicable.

This demonstrates the full pipeline end-to-end without needing the full BIS catalogue.

From <<https://claude.ai/chat/076518c5-2ebe-4e43-94ff-c174d9f04cd3?artifact=194f941d-d0de-4e70-bfe9-915cb892a93e>>